

## Q1 2021 (27 Aug Shift 2)

If  $y(x) = \cot^{-1} \left( \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right)$ ,  $x \in \left( \frac{\pi}{2}, \pi \right)$ ,

then  $\frac{dy}{dx}$  at  $x = \frac{5\pi}{6}$  is:

(1)  $-\frac{1}{2}$

(2)  $-1$

(3)  $\frac{1}{2}$

(4)  $0$

## Q2 2021 (27 Aug Shift 1)

If  $y^{14} + y^{-14} = 2x$ , and  $(x^2 - 1) \frac{d^2y}{dx^2} + \alpha x \frac{dy}{dx} + \beta y = 0$  then  $|\alpha - \beta|$  is equal to .

## Q3 2021 (26 Aug Shift 1)

If  $y = y(x)$  is an implicit function of  $x$  such that  $\log_e(x + y) = 4xy$ , then  $\frac{d^2y}{dx^2}$  at  $x = 0$  is equal to .

### Q3 (40)

Q1 (1)

$$y(x) = \cot^{-1} \left[ \frac{\cos \frac{x}{2} + \sin \frac{x}{2} + \sin \frac{x}{2} - \cos \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2} - \sin \frac{x}{2} + \cos \frac{x}{2}} \right]$$

$$y(x) = \cot^{-1} \left( \tan \frac{x}{2} \right) = \frac{\pi}{2} - \frac{x}{2}$$

$$y'(x) = \frac{-1}{2}$$

Q2 (17)

$$y^{\frac{1}{4}} + \frac{1}{y^{\frac{1}{4}}} = 2x$$

$$\Rightarrow \left(y^{\frac{1}{4}}\right)^2 - 2xy^{\left(\frac{1}{4}\right)} + 1 = 0$$

$$\Rightarrow y^{\frac{1}{4}} = x + \sqrt{x^2 - 1} \text{ or } x - \sqrt{x^2 - 1}$$

$$\text{So, } \frac{1}{4} \cdot \frac{1}{y^{\frac{3}{4}}} \frac{dy}{dx} = 1 + \frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{1}{4} \cdot \frac{1}{y^{\frac{3}{4}}} \frac{dy}{dx} = \frac{y^{\frac{1}{4}}}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{4y}{\sqrt{x^2 - 1}} \dots (1)$$

$$\text{Hence, } \frac{d^2y}{dx^2} = 4 \frac{(\sqrt{x^2 - 1})y' - \frac{yx}{\sqrt{x^2 - 1}}}{x^2 - 1}$$

$$\Rightarrow (x^2 - 1)y'' = 4 \frac{(x^2 - 1)y' - xy}{\sqrt{x^2 - 1}}$$

$$\Rightarrow (x^2 - 1)y'' = 4 \left( \sqrt{x^2 - 1}y' - \frac{xy}{\sqrt{x^2 - 1}} \right)$$

$$\Rightarrow (x^2 - 1)y'' = 4 \left( 4y - \frac{xy'}{4} \right) \text{ (from I)}$$

$$\Rightarrow (x^2 - 1)y'' + xy' - 16y = 0$$

$$\text{So, } |\alpha - \beta| = 17$$

Q3 (40)

$$\ln(x + y) = 4xy \quad (\text{At } x = 0, y = 1)$$

$$x + y = e^{4xy}$$

$$\Rightarrow 1 + \frac{dy}{dx} = e^{4xy} \left( 4x \frac{dy}{dx} + 4y \right)$$

$$\text{At } x = 0 \quad \boxed{\frac{dy}{dx} = 3}$$

$$\frac{d^2y}{dx^2} = e^{4xy} \left( 4x \frac{dy}{dx} + 4y \right)^2 + e^{4xy} \left( 4x \frac{d^2y}{dx^2} + 4y' \right)$$

$$\Rightarrow \frac{d^2y}{dx^2} = 40$$